CSA0613-Design and Analysis of Algorithms

NAME: Safiya B

REG NO: 192411043

Q33 Report – Linear Search in Library System

1. Aim

To implement Linear Search in C to search for a book ID in an unsorted library list and analyze its time complexity.

2. Algorithm

1. Start.
2. Read the number of book IDs (n).
3. Input all book IDs into an array.
4. Read the book ID to be searched.
5. Compare the search key with each book ID one by one.
6. If a match is found, display the position and stop searching.
7. If no match is found after checking all elements, display "Book not found."
8. Stop.

3. Source Code

main.c

```c
#include <stdio.h>

int main()
{
    int n, i, key, found = 0;

    printf("Enter the number of books: ");
    scanf("%d", &n);

    int book[n];

    printf("Enter the Book IDs:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &book[i]);
    }

    printf("Enter the Book ID to search: ");
    scanf("%d", &key);

    for(i = 0; i < n; i++)
    {
        if(book[i] == key)
        {
            found = 1;
            printf("Book found at position %d\n", i + 1);
            break;
        }
    }

    if(found == 0)
    {
```

```c
    printf("Enter the number of books: ");
    scanf("%d", &n);

    int book[n];

    printf("Enter the Book IDs:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &book[i]);
    }

    printf("Enter the Book ID to search: ");
    scanf("%d", &key);

    for(i = 0; i < n; i++)
    {
        if(book[i] == key)
        {
            found = 1;
            printf("Book found at position %d\n", i + 1);
            break;
        }
    }

    if(found == 0)
    {
        printf("Book not found.\n");
    }

    return 0;
}
```

4. Output

```
Enter the number of books: 5
Enter the Book IDs:
101
205
150
320
450
Enter the Book ID to search: 320
Book found at position 4


...Program finished with exit code 0
Press ENTER to exit console.
```

5. Explanation

The program stores book IDs in an unsorted array. It searches for the required book ID using Linear Search, which compares each element sequentially until the book is found or the list ends. Since the data is unsorted, Linear Search is the most suitable searching method.

6. Complexity Analysis

- Best Case: O(1) – The book is found at the first position.
- Average Case: O(n) – The book is found somewhere in the middle.
- Worst Case: O(n) – The book is at the last position or is not present.

7. Conclusion

Linear Search is simple and effective for searching in an unsorted library database. It requires no sorting and works well when books are frequently added or deleted. However, for very large datasets with frequent searches, maintaining a sorted list and using Binary Search can provide faster search performance.